

JUNXIAN WU

Senior AI Engineer | Applied AI, Agentic Systems & Production Delivery

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PROFILE

Senior AI Engineer who takes AI features from product requirements and prototypes through model/runtime integration, evaluation, release gates, and production handoff. Production depth in computer vision, edge AI, and native C++/Android; open-source work extends that discipline into AI-agent governance, orchestration, and verifiable delivery. Strong at debugging constrained systems and aligning product, runtime, QA, and customer-facing teams.

Agents & evaluation: agent governance, capability-aware routing, retrieval, enforced gates, human review, deterministic traces, hard-case evaluation
ML & runtime: Python, C++, PyTorch, TensorFlow, OpenCV, Android/NDK, TFLite, ONNX, TensorRT, OpenGL/GLSL
Delivery: quantization, export validation, device debugging, Docker, FastAPI/Flask, acceptance criteria, production handoff

PROFESSIONAL EXPERIENCE

Senior AI Engineer | Black Sesame Technologies, Singapore

Jan 2025 - Present

- Own end-to-end delivery for camera and image AI features, translating product requirements and field feedback into technical scope, prototypes, hard-case evaluation, release gates, and maintainable handoff.
- Coordinate product, camera tuning, runtime, QA, and customer-facing teams around reproducible issues, acceptance decisions, on-device debugging, and release readiness.

AI Engineer | Black Sesame Technologies, Singapore

Aug 2022 - Jan 2025

- Productized models for portrait segmentation and matting, monocular depth, object detection, and camera-oriented image processing.
- Built and maintained training-to-deployment and native-integration workflows across quantization, export validation, C++/Android pipelines, device deployment, and release review.

SELECTED PROJECTS

Real-time Composition & Portrait Imaging | Detection, tracking, segmentation, depth

Company project

- Led a real-time camera AI system combining object detection, optical-flow tracking, segmentation and matting, monocular depth, GPU rendering, and application-state behavior.
- Drove hard-case evaluation, on-device debugging, acceptance decisions, and release readiness with product, runtime, camera tuning, QA, and customer-facing teams.
- Converted field issues into reproducible scenarios and balanced visual quality, latency, device constraints, and stable behavior across camera state transitions.

Depth-aware Bokeh & Portrait Rendering | Monocular depth, C++, Android, GPU

Company project

- Developed a depth-aware portrait rendering workflow from model preparation and export validation through native integration and device-side visual debugging.
- Built hard-case datasets, batch comparison, golden cases, and release-review tooling to diagnose long-tail visual failures and support model replacement decisions.
- Integrated depth outputs into C++/Android camera pipelines and compared model changes through side-by-side and golden-output reviews across challenging subjects and scenes.

Model-to-Device AI Delivery Workflow | PyTorch, quantization, TFLite/ONNX

Company workflow

- Built a repeatable training-to-deployment workflow spanning data preparation, quantization, export validation, runtime assumptions, native C++/Android integration, and device deployment.
- Applied the workflow across portrait segmentation and matting, monocular depth, object detection, and camera-oriented image processing, with explicit handoff and release checks.
- Standardized simulation, cross-platform builds, runtime-compatibility checks, device deployment, and visual review so regressions could be reproduced earlier in the release cycle.

VEMO - AI Agent Governance Framework | Python, YAML, Git/CI

Open source

- Designed and built a vendor-neutral governance framework for AI coding agents with risk-tiered task control, capability-aware routing, durable repository state, and just-in-time specification loading.
- Enforced scope, secret, run-budget, acceptance, and independent-review gates across agent hooks, Git pre-commit/pre-push, and server-side CI.
- Delivered a zero-dependency Python CLI, stack presets, fleet management, and executable conformance evaluation with 68/68 validator and hook end-to-end checks. [Code and architecture](#)

RESEARCH, EDUCATION & PUBLICATION

Research Intern, National University of Singapore | Dec 2021 - Jul 2022 | Computer-vision research for construction-site safety; co-authored a peer-reviewed paper.

Research Assistant, Shandong University | Sep 2019 - Jun 2021 | Applied computer vision and machine learning to interdisciplinary datasets, experiments, visualization, and reporting.

Master of Technology, Intelligent Systems | NUS-ISS | 2021 - 2022 | GPA 4.33 / 5.00

Bachelor of Engineering, Software Engineering | Shandong University | 2017 - 2021 | GPA 86.86 / 100 | Outstanding Graduate

Publication: Co-author, "Automated classification of 'cluttered' construction housekeeping images through supervised and self-supervised feature representation learning," *Automation in Construction*, 2023. [DOI](#)